

Answer Key & Teacher Notes

Session 1 – Customary Unit Conversions • Session 2 – Metric Unit Conversions

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Customary Unit Conversions

TALK PROMPTS

- When you convert larger units (feet) to smaller units (inches), do you multiply or divide? How do you know?
- How many inches are in 1 yard? Use that number to check the gym rope total a second way.

PARTNER TASK

Water Cooler, Party Punch, Gym Rope. One partner converts one fact at a time; the other uses a combined fact (1 yd = 36 in, 1 gal = 16 cups, 1 gal = 8 pt) to check the total. Switch roles for each problem.

WATCH FOR

- Multiplying when changing to a bigger unit (60 in = 720 ft).
- Using a wrong fact, such as 1 ft = 10 in or 1 qt = 2 cups.
- Using the wrong fact to change feet to yards (24 ÷ 12 instead of 24 ÷ 3 for the garden fence).

SESSION 2 • TEACHER NOTES

Metric Unit Conversions

TALK PROMPTS

- Why is converting in the metric system so fast compared to customary units?
- Why did the number get bigger when you changed meters to centimeters?

PARTNER TASK

5K Run, Lab Mass, Lab Ribbon. One partner names the operation and the power of 10 (× 1,000, ÷ 100, ...); the other converts and then checks by converting back. Switch roles for each problem.

WATCH FOR

- Dividing when changing to a smaller unit (3 m = 0.03 cm).
- Using the wrong power of 10, such as 1 km = 100 m.
- Dividing amounts in different units (2.4 m ÷ 30 cm) before making the units match.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.3d	1	1	1	1	2	2	1	3	3	1	1
S1 • Form B	6.RP.A.3d	1	1	1	1	2	2	1	3	3	1	1
S2 • Form A	6.RP.A.3d	1	1	1	1	2	2	1	3	3	1	1
S2 • Form B	6.RP.A.3d	1	1	1	1	2	2	1	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

customary units: U.S. units like inches, **pounds**, cups and gallons.

convert: Write the same amount with a **different** unit.

conversion factor: A fact that shows two **equal** amounts.

- Big unit to small unit: **multiply**. The number gets bigger.
- Small unit to big unit: **divide**. The number gets smaller.

Example: Quarts to cups: $2 \times 4 = 8$ cups. Cups to fluid ounces: $8 \times 8 = 64$ fl oz. Check backward: $64 \div 8 = 8$ cups, and $8 \div 4 = 2$ quarts. Answer: 2 quarts = **64** fluid ounces.

YOU TRY

1. a. 9 ft b. 2 lb
2. $7 \div 3 = 2 \frac{1}{3}$ yards, so buy 3 yards

EXIT TICKET

1. a. 72 in b. 4 gal
2. $2.5 \times 16 = 40$ oz
3. **A** (multiply)

PRACTICE FORM A

1. a. 48 in b. 15 ft c. 72 oz d. 5 gal e. 3.5 lb f. 12 qt
2. 4 ft = 48 in. Yes, he is exactly 48 inches, so he can ride.
3. a. $2 \times 4 = 8$ qt b. $8 \times 4 = 32$ cups
4. a. $8 \times 3 = 24$ ft b. $24 \times 12 = 288$ in
5. $2.5 \times 5,280 = 13,200$ feet
6. $9 \times 4 = 36$ qt; $36 \times 2 = 72$ pints, so 72 cups
7. $24 \div 4 = 6$ gallons
8. a. $24 \div 3 = 8$ yd b. $8 \times \$9 = \72 c. Yes. $24 \times \$3.25 = \78 , which is \$6 more.
9. Feet are bigger than inches, so the number must get smaller; Sam should divide. $60 \div 12 = 5$ feet.
10. **C** (60)
11. **D** (3)

Self-check word INCHES

PRACTICE FORM B

1. a. 60 in b. 21 ft c. 56 oz d. 7 gal e. 2.5 lb f. 24 qt
2. 5 ft = 60 in. Yes, 60 inches is more than 54 inches.
3. a. $4 \times 4 = 16$ qt b. $16 \times 4 = 64$ cups
4. a. $6 \times 3 = 18$ ft b. $18 \times 12 = 216$ in
5. $1.5 \times 5,280 = 7,920$ feet
6. $7 \times 4 = 28$ qt; $28 \times 2 = 56$ pints, so 56 cups
7. $36 \div 4 = 9$ gallons
8. a. $30 \div 3 = 10$ yd b. $10 \times \$8 = \80 c. Yes. $30 \times \$2.90 = \87 , which is \$7 more.
9. Feet are bigger than inches, so the number must get smaller; Maya should divide. $84 \div 12 = 7$ feet.
10. **C** (48)
11. **A** (4)

Self-check word POUNDS

GUIDED NOTES

metric units: Meters, grams and liters, built on **tens**.

kilo- / centi- / milli-: Kilo means **1,000** times. Centi means one hundredth. Milli means one thousandth.

powers of 10: 10, 100, 1,000. Metric changes **shift** the decimal point.

- Big unit to small unit: **multiply** by 10, 100 or 1,000.
Small to big: divide.
- Make the units **match** before you divide or compare.

Example: Make the units match: $2.4 \times 1,000 = \mathbf{2,400}$ g.
Divide: $2,400 \div 300 = \mathbf{8}$. Check: $8 \times 300 = 2,400$ g.
Answer: You can make **8** portions.

YOU TRY

1. a. 320 cm b. 0.75 kg
2. $15 \times 1,000 = 15,000$ mL; $15,000 \div 750 = 20$ jugs

EXIT TICKET

1. a. 6,500 m b. 0.8 m
2. $2 \times 1,000 = 2,000$ mL; $2,000 \div 250 = 8$ cups
3. **C** (400 g)

PRACTICE FORM A

1. a. 7,000 m b. 1.5 L c. 3,000 g d. 2.5 m e. 2.45 km f. 0.0025 L
2. $4 \text{ L} = 4,000 \text{ mL}$; $4,000 \div 500 = 8$ bottles
3. a. $5 \times 1,000 = 5,000$ m b. $5,000 - 3,200 = 1,800$ m
4. a. $350 \times 1,000 = 350,000$ mg b. $350 \div 1,000 = 0.35$ kg
5. a. $2.4 \times 100 = 240$ cm b. $240 \div 30 = 8$ pieces
6. $4,800 \div 1,000 = 4.8$ kg. Yes, 4.8 kg is under 5 kg.
7. $12.75 \times 1,000 = 12,750$ g
8. a. $8 \times 450 = 3,600$ m b. $3,600 \div 1,000 = 3.6$ km c. Sample: kilometers, because 3.6 km is a small, easy number for a long distance.
9. Centimeters are smaller than meters, so the number must get bigger; Daniel should multiply. $3 \times 100 = 300$ cm.
10. **A** (4,000)
11. **D** (10)

Self-check word DECIMAL

PRACTICE FORM B

1. a. 9,000 m b. 2.5 L c. 6,000 g d. 1.75 m e. 3.82 km f. 0.0045 L
2. $6 \text{ L} = 6,000 \text{ mL}$; $6,000 \div 250 = 24$ cups
3. a. $6 \times 1,000 = 6,000$ m b. $6,000 - 2,700 = 3,300$ m
4. a. $425 \times 1,000 = 425,000$ mg b. $425 \div 1,000 = 0.425$ kg
5. a. $2.4 \times 100 = 240$ cm b. $240 \div 40 = 6$ pieces
6. $5,300 \div 1,000 = 5.3$ kg. No, 5.3 kg is over 5 kg.
7. $9.6 \times 1,000 = 9,600$ g
8. a. $6 \times 750 = 4,500$ m b. $4,500 \div 1,000 = 4.5$ km c. Sample: kilometers, because 4.5 km is a small, easy number for a long distance.
9. Grams are smaller than kilograms, so the number must get bigger; Maya should multiply. $5 \times 1,000 = 5,000$ g.
10. **B** (2.5)
11. **C** (3,000)

Self-check word METRICS

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • CUSTOMARY UNIT CONVERSIONS

1. a. 96 in b. 7 yd c. 80 oz d. 3 gal e. 14 cups f. 2.5 lb
2. a. $<$ (3 ft = 36 in) b. $<$ (5 pt = 2.5 qt) c. $>$ (2 lb = 32 oz) d. = (3 yd = 9 ft)
3. 3 gal = 12 qt; 20 qt = 5 gal; 7.5 gal = 30 qt
4. 3 gal = 12 qt = 24 pt = 48 cups
5. No. 2 yd = $2 \times 3 = 6$ ft = $6 \times 12 = 72$ in. 70 in is less than 72 in, so she is 2 inches short.

SHEET 2 • METRIC UNIT CONVERSIONS

1. a. 3,500 m b. 0.85 m c. 6,000 mL d. 0.45 kg
2. a. $18 \times 1,000 = 18,000$ mL b. $18,000 \div 300 = 60$ cups
3. a. $4 \times 800 = 3,200$ m b. $3,200 \div 1,000 = 3.2$ km
4. a. $3 \times 400 = 1,200$ g b. No. 1.5 kg = 1,500 g; $1,500 - 1,200 = 300$ g short
5. He divided by 1,000 instead of multiplying.
Kilograms to grams is large to small, so multiply: $4.5 \times 1,000 = 4,500$ g.
6. **B** (160 mm)
7. **C** (600 mL)